

Particulate Matter Monitoring & Health Effects

Issue 2026-09-14 | PubMed entry window 14 September 2026 | 16 records screened in scope

16

RECORDS IN SCOPE
(11 EXCLUDED, 6 HELD)

6

OF 10 SUBTOPIC
BANDS POPULATED

5

SENSING RECORDS
(31% OF CORPUS)

4

EFFECT ESTIMATES
(3 WITH A CI, 0 ON PM)

8

RECORDS CARRYING
NO PMID

Signal of the day

Withdrawing a monitor changed the pollution, not just the record of it.

Zhou et al. (PNAS) treat the 2025 suspension of the US embassy air-quality programme as a natural experiment and ask what happens to a city when a trusted, independent monitoring node goes dark. Using real-time satellite AOD in a matched difference-in-differences design against same-country control cities, they find **no differential pre-trend** across 46 cities and then a prompt **24–33% rise in local pollution** once reporting stops, largest and most persistent where no substitute data source existed. Calibrated against concurrent pre-shutdown embassy PM_{2.5}, the preferred conservative translation is **+1.9 µg/m³**; alternative translations span 1.5 to 16.3 µg/m³, and no confidence interval is reported for the headline effect — so the causal design is considerably stronger than the magnitude. The result inverts the usual framing of this watch. Dense low-cost networks are normally justified as a way to *observe* exposure at a resolution regulatory networks cannot reach; here, observation is itself the intervention, and its withdrawal is measurable in the atmosphere. Whatever a distributed network costs, this paper is the argument that its governance and continuity matter as much as its calibration.

What changed today

- ▶ **A thin day on the health axis, and the thinness is topical.** Registry-wide PubMed deposits for the 14 September entry date stand at **2,375**, against 4,593 for 13 September. Both PubMed queries together returned eight candidates; two were screened out and none reported a new PM exposure–response estimate. Four of the ten subtopic bands — *Neuro*, *Cardiovascular & metabolic*, *Reproductive & developmental* and *Other clinical* — are empty, the eighth issue of fifty to populate exactly six.
- ▶ **Instrumentation carried the issue, and PubMed indexed almost none of it.** Five of sixteen records are sensing or instrumentation work; **eight of sixteen records carry no PMID at all**, arriving through the Crossref-by-ISSN leg (*AMT*, *ACP*, *Atmosphere*, *EST*) or the Consensus sweep. A PubMed-only watch would have shipped a six-record issue today.
- ▶ **Regional, not universal, is now the settled answer on low-cost correction.** Gupta et al. conclude from PA-II co-locations in Raleigh and Delhi that location-specific pre-deployment calibration is a prerequisite for scientific use, and Gökozan's ablation study shows that where continuous co-location is not feasible, humidity-interaction and cyclic-time features recover most of the gap (test R^2 0.724 → 0.847) but leave an RMSE of 5.19 µg/m³ — large against typical ambient means. The two papers agree on the diagnosis and differ only on the fallback.
- ▶ **No PM exposure estimate with a confidence interval was published into this window.** The three interval-bearing estimates in today's forest plot are all *heat* exposure, from the COPD meta-analysis; the fourth, the embassy effect, has no interval. This is the thirteenth issue in fifty with no PM ratio estimate carrying a CI, so the figure should be read as a record of what was available, not as evidence about PM.
- ▶ **Two records were removed by the DOI gate after title screening passed them.** Both Consensus hits on PurpleAir transferability and Bogotá LCS calibration had already shipped — on 29 July and 3 August — under *rewritten* titles, one a paraphrase and one entirely different. Title-normalised deduplication is structurally incapable of catching this; only the DOI check is.
- ▶ **Six in-scope records were verified and deliberately not shipped.** Their DOIs resolve, but no abstract exists in Crossref, Europe PMC, Scholar Gateway or Consensus, and the publisher pages are client-rendered. The priority pickup is the *Aerosol Science and Technology* Spider-MAGIC and OPS size-distribution evaluation — a tier-A instrument paper held only for want of 200 words.

Corpus analytics

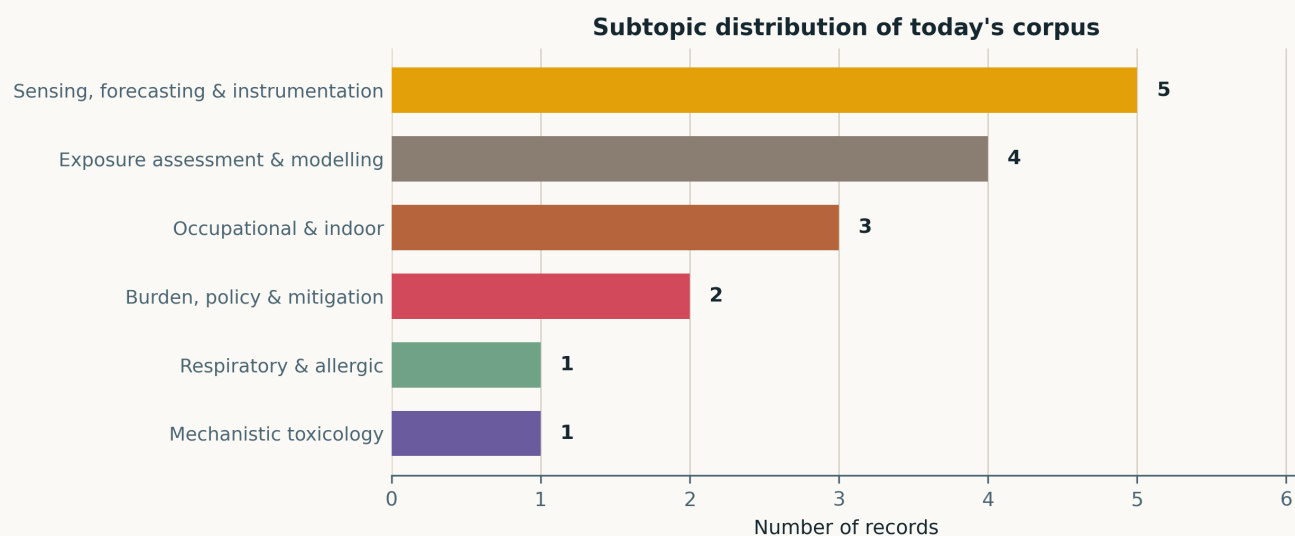


Fig. Subtopic assignment of the 16 in-scope records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* leads at 5, followed by *Exposure assessment & modelling* at 4 and *Occupational & indoor* at 3. Only six of the ten bands are populated; the four clinical-epidemiology bands that usually carry a daily issue are all empty.

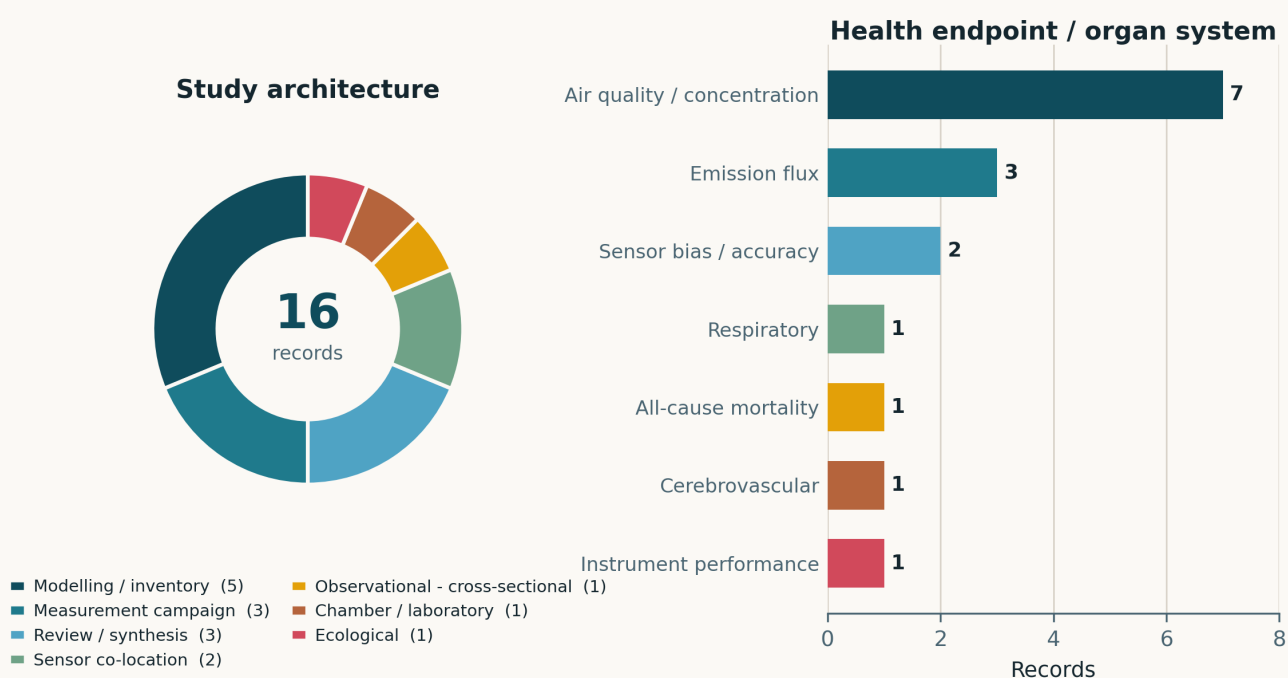


Fig. Left — study architecture. Modelling and inventory work is the largest class (5), with measurement campaigns and reviews at 3 each and sensor co-location at 2; a single record is observational cross-sectional and a single one chamber/laboratory. Sixteen design labels were coined for this issue and added to the classifier in the same edit, because the existing map was built for clinical epidemiology and this corpus is not. Right — endpoint. **Thirteen of sixteen records carry no human health endpoint:** seven report air-quality concentrations, three emission flux, and three sensor or instrument performance. Three clinical endpoints appear in total, one each — respiratory, cerebrovascular and all-cause mortality.

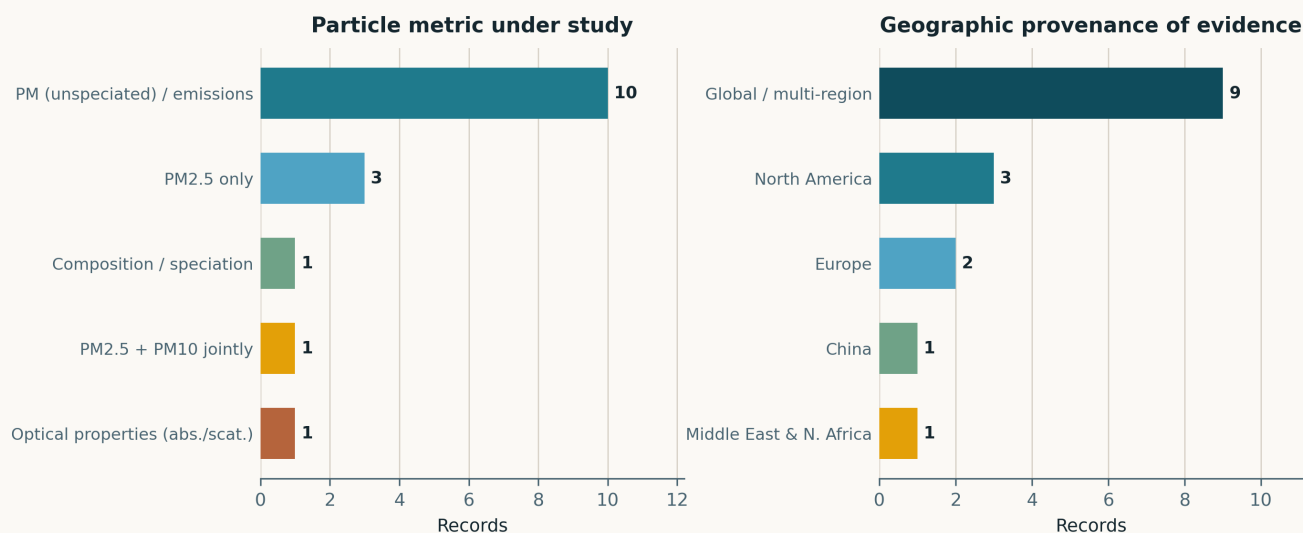


Fig. Left — particle metric. Unspeciated PM or emissions is the exposure in 10 of 16 records, PM_{2.5} alone in 3; composition/speciation and optical properties appear once each. Right — geographic provenance. Nine records chart as global or multi-region, which reflects the day's composition rather than breadth: burden syntheses, satellite products and multi-continent syntheses land there, and the purely numerical sMCPC study folds into the same bucket for want of a geography. The Raleigh-and-Delhi co-location spans two regions and is charted under one; China contributes a single record.

Life-course windows addressed today (records may span several stages)

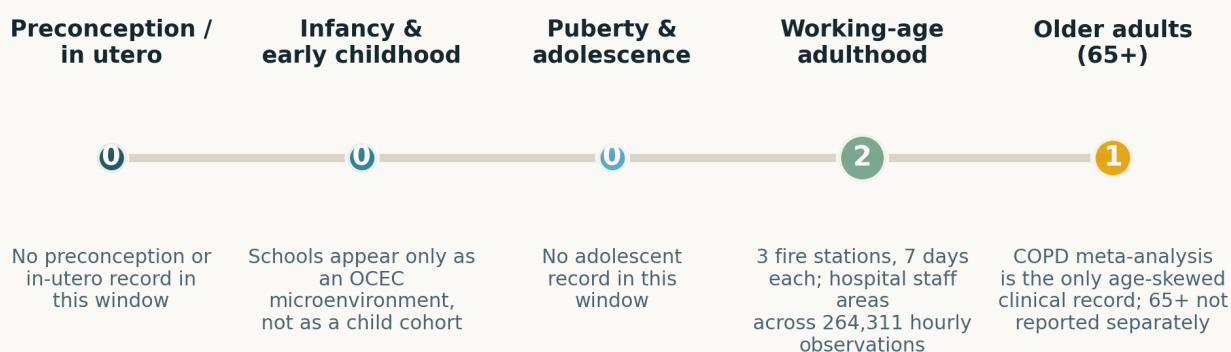


Fig. Life-course windows addressed. **Three of the five stages are empty today** — no preconception or in-utero, infancy or adolescent record appears at all. The working-age count reflects two occupational monitoring deployments (three fire stations; hospital staff areas), not a health-outcome cohort, and the 65+ entry is the COPD meta-analysis, which does not report an age-stratified estimate.

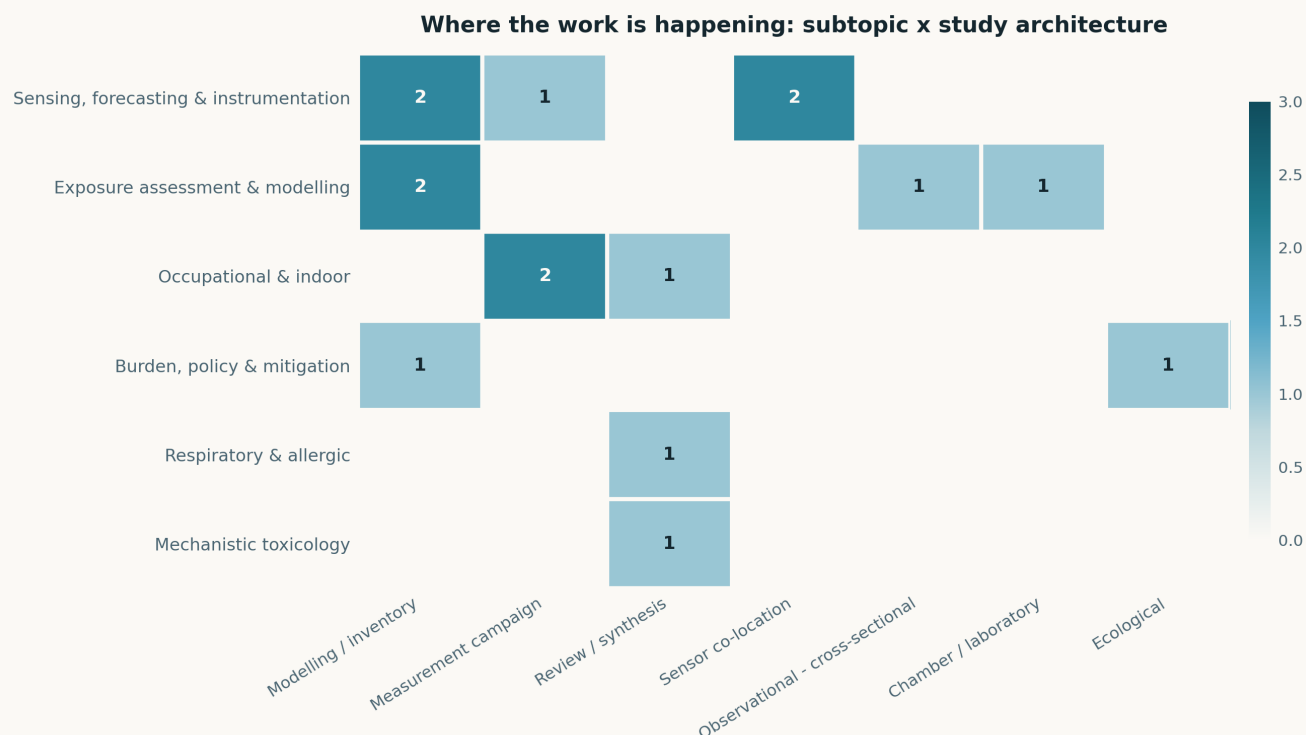


Fig. Subtopic × study-architecture cross-tabulation. The grid is sparse by construction at $n = 16$, and the structure it shows is the day's defining feature: the sensor co-location cells sit entirely inside *Sensing*, the modelling cells spread across *Sensing* (2), *Exposure assessment* (2) and *Burden* (1), and no cell links an instrumentation architecture to a clinical band — the two clinical rows each hold a single review. Instrumentation and epidemiology did not meet anywhere in this corpus.

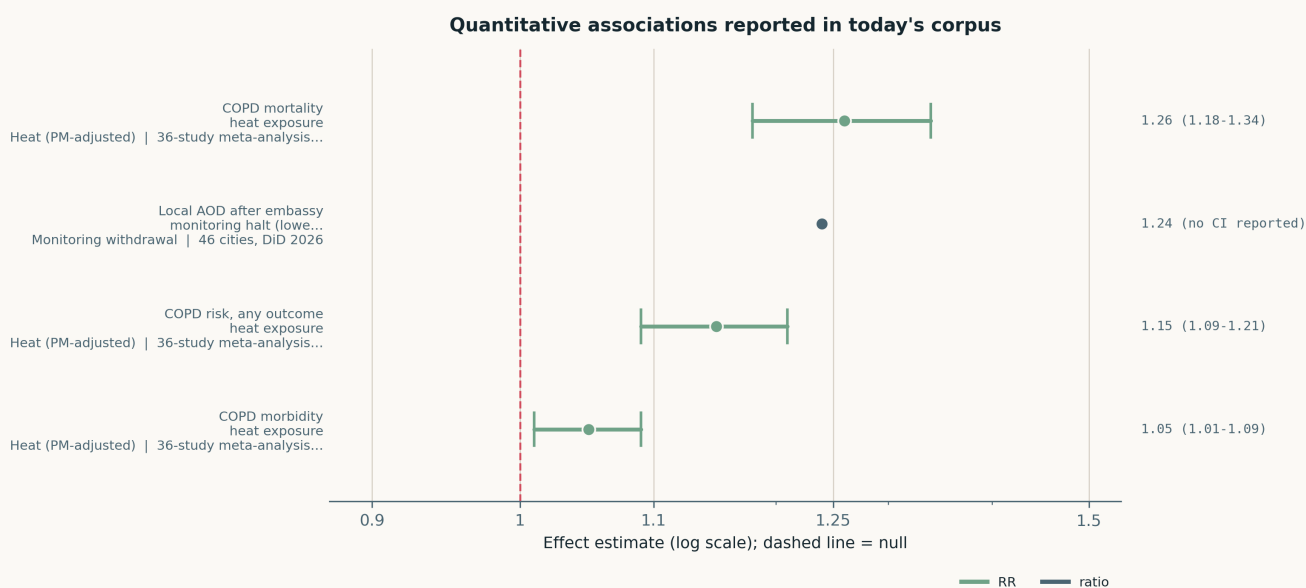


Fig. All quantitative effect estimates in the corpus on a common logarithmic axis — four, which is few, and **none of them is a particulate exposure**. Three are *heat* exposure from the 36-study COPD meta-analysis, shown here because the air-pollution-adjusted sensitivity analysis is what makes them relevant to this watch; the mortality and morbidity estimates are strata of the overall estimate and are not independent of it. The fourth is the lower bound of the 24–33% embassy-shutdown range, plotted without an interval because none was reported. Read this figure as a statement about what the window contained, not about PM risk.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

5 records

Zhou et al. 2026 Proc Natl Acad Sci U S A

Air pollution after the shutdown of US embassy monitoring

Matched difference-in-differences across 46 cities with same-country controls, using real-time satellite AOD. No differential pre-trend; a prompt **24–33% rise in local pollution** after US reporting stopped, largest where substitutes were poorest. Calibrating AOD to concurrent pre-shutdown embassy PM_{2.5} gives **+1.9 µg/m³** on the authors' conservative translation, with alternatives spanning 1.5–16.3 — an order-of-magnitude spread that sits in the calibration step, not the DiD, and no CI accompanies the headline. The design is the contribution: monitoring treated as an intervention with a measurable atmospheric effect.

doi: 10.1073/pnas.2605467123

Zhou et al. 2026 Atmos Meas Tech

Performance modeling of a small mixing-type condensation particle counter

Validated CFD model of the sMCPC particle channel resolving turbulent flow, temperature and species transport, with Kelvin, non-continuum and latent-heat terms in the growth calculation. Raising the saturator-cooler temperature difference ΔT raises supersaturation and lowers the critical activation diameter $D_{p,50}$; raising total flow Q_g suppresses supersaturation and shifts its peak downstream; both ΔT and vapour fraction steepen the $D_{p,50}$ slope. Everything is numerical — counting efficiency is asserted through one validation case, not measured across the swept space. This is design-space guidance of the kind a portable ultrafine network node needs before it goes to the field.

doi: 10.5194/amt-19-5815-2026

Gupta et al. 2026 Aerosol Air Qual Res

Evaluating and calibrating low-cost air quality sensors in contrasting aerosol regimes

PurpleAir PA-II co-located with FEM monitors for 2–4 weeks in Raleigh, North Carolina and Delhi, India, deliberately spanning a clean and a heavily loaded regime. A model on PA PM_{2.5}, temperature and RH holds mean absolute bias within **12% hourly and 6% daily**. The headline is negative: pre-deployment calibration specific to location or region is a prerequisite for scientific use, i.e. no single transferable correction exists. Two to four weeks is short against the seasonal cycle in either city, so drift and seasonal composition change lie outside what this design can see. *Connector-sourced; outside the 14 September entry window.*

doi: 10.1007/s44408-026-00146-x

Gökozan 2026 Chemosensors

Machine-learning calibration of a low-cost PM_{2.5} sensor with location-specific covariates and feature engineering

Ablation design separating environmental covariates, auxiliary gases and feature engineering for one outdoor residential sensor, validated on a leakage-safe time-based split. Feature engineering helped in every configuration: raw PM_{2.5} plus engineered features raised Random Forest test R^2 from **0.724 to 0.847** and cut RMSE from 6.96 to **5.19 µg/m³**; humidity interactions and cyclic time encodings carried the importance. One sensor at one site is the limit, and 5.19 µg/m³ is large against typical ambient means — but the operational claim, that useful correction is reachable without continuous co-location, is the one network operators need tested. *Connector-sourced; outside the entry window.*

doi: 10.3390/chemosensors14070154

Sun et al. 2026 Atmos Chem Phys

Aerosol properties during a transatlantic smoke transport event from Aeolus observations

ALADIN, a spaceborne high-spectral-resolution lidar, retrieves extinction and backscatter directly rather than through an assumed lidar ratio, and is used to follow a September 2020 western-US smoke plume to Europe. Mean extinction near 200 Mm⁻¹ across selected cross-sections; AOD fell 0.54 → 0.33 along the path; the layer rose from ~4.4 km over the US to above 6 km over Europe and arrived above cloud, with lidar ratios first falling then rising. One event constrains generality. For satellite-to-surface PM_{2.5} fusion the relevant point is blunt: this plume was invisible at ground level throughout.

doi: 10.5194/acp-26-12953-2026

Exposure assessment & modelling

4 records

Ajibade et al. 2026 Proc Natl Acad Sci U S A

Gentrification, displacement, disadvantage and exclusivity predict urban environmental and health inequities

PCA on 2010 and 2020 ACS and Census data yields four contemporary segregation typologies for 342 Portland tracts, compared against historical HOLC zones across 104 environmental and health variables. HOLC grade predicted persistent exclusivity in the urban core but not injustice elsewhere; **persistently disadvantaged tracts, often outside HOLC boundaries entirely**, carried the highest pollution and hazard exposure and the worst health and disability outcomes. Exclusive neighbourhoods had more canopy and better preventive care *and* more air pollution — an awkward result the paper does not resolve. Wholly ecological and cross-sectional. For monitor siting the message is that a single historical redlining layer is an inadequate stratifier.

doi: 10.1073/pnas.2422760122

Lai et al. 2026 Atmosphere

Monte Carlo model for wind-driven PM emissions from industrial granular materials

First quantitative evaluation of a saltation-and-sandblasting emission model for wind erosion of granular stockpiles, with the Monte Carlo impact count fixed rather than run to convergence. Applied to lead and zinc sulphide concentrates at a Sardinian plant, cutting simulated impacts from **10,000 to 500** changed the median emission by a small relative difference at a twentyfold saving, and the revised model reproduced emission magnitude, sandblasting efficiency and their friction-velocity dependence. Validation is internal and physical-consistency based — no measured emission flux is compared against, which is the central weakness. Fugitive stockpile emission is exactly the source class regulatory networks miss and fenceline arrays are asked to capture.

doi: 10.3390/atmos17090891

Groenen et al. 2026 Sci Total Environ*Characteristics and trend of wildfire activity in the Southern Hemisphere*

Two decades of MODIS active-fire detections across Southern Africa, South America and Oceania. Seasonality is strongly heterogeneous: Southern Africa shows a stable 20-week May–October season with low inter-annual variability, South America a 14-week July–October season with high variability, and **Oceania no clear season at all**, which the authors read as co-existing fire regimes. Active-fire counts are a detection proxy, not emitted mass — insensitive to smouldering, biased by overpass timing and cloud — so no PM emission inventory follows directly. Note that the abstract's stated period (2004–2024) disagrees with the title's (2003–2022); check the paper before citing either.

doi: 10.1016/j.scitotenv.2026.182240

Shukla et al. 2026 Atmos Chem Phys*Organic vapors from savannah and boreal fire emissions under photochemical and dark aging*

Controlled burns of understudied African savannah and European boreal fuels sampled by high-resolution PTR-MS, then aged photochemically and in the dark in a 29 m³ Teflon chamber. Total primary organic-vapour emission factors spanned **69–161 g/kg** by fuel type. Photochemical ageing depleted furanics, phenolics and oxygenated aromatics and enriched carbonyls and small O-containing species; dark low-NO_x ageing produced almost no compositional change, and clustering attributed **73% of group-level compositional variance to oxidation regime**. Chamber burns of prepared fuels do not reproduce field moisture, combustion phase or dilution, so these are upper-bound laboratory factors. Any smoke PM_{2.5} forecast assuming a fixed downwind secondary yield is contradicted here.

doi: 10.5194/acp-26-12977-2026

Occupational & indoor exposure**3 records****Bellacov et al. 2026** J Occup Environ Hyg*Diesel-related pollutant migration in fire stations measured with low-cost direct-reading monitors*

Seven consecutive days of minute-level monitoring at fixed engine-bay and adjacent occupied locations in three fire stations, aggregated to daily means and fitted by hierarchical mixed-effects models. Coarse PM, NO₂, VOCs and CO₂ were significantly associated with engine activity, source distance and station identity, but the *direction* varied by pollutant — and **PM_{2.5} and PM₁ were not significantly associated with any of them**. Measurable concentrations beyond the bays indicate migration or shared indoor sources; this design cannot separate them. Daily aggregation of minute data discards precisely the transient peaks that dominate diesel exposure, so the null on the fine fractions reads as a resolution artefact rather than an absence.

doi: 10.1080/15459624.2026.2722264

Fermoso et al. 2026 Atmosphere*Integrated hospital IAQ monitoring: continuous sensing with targeted chemical characterisation*

Thirty months of continuous low-cost sensor monitoring across hospital microenvironments (June 2023 – December 2025) yielding **1.79 million raw records, 1.42 million retained indoor measurements and 264,311 hourly aggregates**, paired with targeted VOC and aldehyde sampling. Distinct recurrent signatures emerged per area: CO₂ tracked occupancy and ventilation demand, while formaldehyde response was persistent and seasonal in a pattern consistent with indoor sources plus ventilation heterogeneity. The formaldehyde inference rests on a low-cost channel with known cross-sensitivity, and the campaign sampling is too sparse to validate it continuously. The **79% retained fraction** is the number to take away when sizing a long-horizon deployment.

doi: 10.3390/atmos17090895

Yang et al. 2026 Environ Sci Technol*Organic contaminants of emerging concern in indoor air: global distribution and hazard burden*

Seven OCEC classes spanning over 130 compounds synthesised across residences, offices, schools and commercial spaces. Microenvironments separate cleanly — phthalate alternatives dominate residential air to 2.08×10^4 ng/m³, PFAS schools to 230 ng/m³, organophosphate esters offices to 159 ng/m³ — and the US, Germany, Sweden and China rank highest on inhalation hazard, which the authors say plainly may reflect monitoring intensity rather than contamination. **Microplastics were excluded from the hazard index because units and toxicity metrics are irreconcilable**: a candid admission that the particulate fraction of indoor emerging contaminants cannot currently be placed on a common scale with the rest.

doi: 10.1021/acs.est.6c06332

Burden, policy & mitigation**2 records****Jiang et al. 2026** Med Gas Res*Ambient particulate matter pollution and stroke burden*

GBD 2021 stroke deaths and DALYs attributable to ambient particulate matter, decomposed by region, sex and SDI over 1990–2021 with EAPC trends and concentration and slope indices of inequality. Absolute deaths rose **107%** and DALYs **101%** while age-standardised rates fell — the familiar decomposition of ageing and population growth against declining risk. Middle-SDI locations carry **46% of the global burden** and exceed high-SDI on both age-standardised measures. The concentration index suggests improving relative inequality while the slope index worsened against 1990, and the paper does not reconcile the two. Attributable burden is a modelled quantity resting on the exposure surface and the chosen CRF; this is not independent epidemiological evidence.

doi: 10.4103/mgr.MEDGASRES-D-25-00188

Liu et al. 2026 Environ Monit Assess*Spatiotemporal evolution of PM_{2.5} and its health and economic effects in Central China*

Henan, Hubei and Hunan analysed with global Moran's I, centroid migration and BenMAP, with Henan carried forward for valuation over 2017–2023. Spatial clustering is strong and stable (**Moran's I 0.524–0.738, all $p < 0.001$**), high-value clusters sit in Henan and around Wuhan and Changsha, and the concentration centroid moved only **2.86–24.37 km** per year. The authors explicitly *reject* a persistent southwestward migration because most displacements fall below their noise threshold — a welcome negative result in a literature prone to over-reading centroid drift. Attributable premature deaths declined with a late rebound, but a rising value of a statistical life meant monetised benefit grew more slowly than health benefit. BenMAP output inherits its CRF and baseline-rate assumptions entirely.

doi: 10.1007/s10661-026-15911-z

Respiratory & allergic

1 record

Kim et al. 2026 Int J Environ Health Res*Heat exposure and COPD morbidity and mortality: systematic review and meta-analysis*

Thirty-six observational studies pooled in a three-level random-effects model that handles non-independent estimates. Heat raised overall COPD risk (RR 1.15, 95% CI 1.09–1.21), with mortality (1.26, 1.18–1.34) well above morbidity (1.05, 1.01–1.09). Heat-wave and threshold definitions both gave RR 1.21 against 1.05 for continuous temperature metrics, and the continuous metrics were also the least heterogeneous — so definitional choice, not biology, drives much of the spread. The finding that earns this record its place here is negative: **adjusting for air pollution did not substantially alter the associations**, which argues against PM confounding of heat–COPD associations while saying nothing about interaction. No publication-bias or exposure-misclassification assessment is reported in the abstract.

doi: 10.1080/09603123.2026.2734444

Mechanistic toxicology

1 record

Vahdatbin et al. 2026 iScience*Is recycling sustainable: the promise and risks of scrap tire valorization*

Roughly 1.5 billion end-of-life tires accumulate annually, and this review argues that devulcanization, pyrolysis and mechanical grinding each release hazardous species including VOCs, PAHs, heavy metals, microplastics and 6PPD-quinone. It is qualitative: no emission factors, exposure concentrations or dose–response relationships are given, so the risk claim is directional and cannot rank the three routes. Its value here is the explicit call for high-fidelity monitoring of recycling facilities — a fenceline low-cost-network application with essentially no published deployment literature. Tire wear is already a leading non-exhaust PM source; tire *reprocessing* as a second, point-source pathway is comparatively unmeasured.

doi: 10.1016/j.isci.2026.117462

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Heat-COPD meta, 36 studies	Heat exposure and its impact on COPD morbidity and mortality: a systematic review and meta-analysis.	<i>Int J Environ Health Res</i>	Systematic review and three-level random-effects meta-analysis	Co-exposure only: air-pollution adjustment tested as a moderator	Global / multi-region
Mechanistic toxicology					
Scrap tire valorization risks	Is recycling sustainable: The promise and risks of scrap tire valorization.	<i>iScience</i>	Narrative review, technical and toxicological	Process emissions: VOCs, PAHs, heavy metals, microplastics, 6PPD-quinone	Global / multi-region
Occupational & indoor					
Fire-station diesel, 3 stations	Characterization of diesel-related pollutant migration in fire stations using low-cost direct-reading monitors.	<i>J Occup Environ Hyg</i>	Repeated-measures monitoring, mixed-effects models	Minute-level PM size fractions, NO ₂ , VOCs and CO ₂ from direct-reading low-cost monitors, 7 days per station	United States
Hospital IAQ, 1.79M records 2023-2025	Toward Integrated Hospital IAQ Monitoring: Continuous Sensing and Targeted Chemical Characterization.	<i>Atmosphere</i>	Long-term continuous sensing plus targeted chemical sampling	CO ₂ , PM _{2.5} , PM ₁₀ , formaldehyde, T, RH, TVOC from low-cost sensors; June 2023 - December 2025	Europe
Indoor OCEC hazard synthesis	Organic Contaminants of Emerging Concern (OCECs) in Indoor Air: A Perspective on Global Distribution and Hazard Burden.	<i>Environ Sci Technol</i>	Systematic synthesis with inhalation hazard prioritisation	Airborne concentrations of 130+ compounds across seven OCEC classes	Global / multi-region
Sensing, forecasting & instrumentation					
Aeolus transatlantic smoke, Sep	Characterization of aerosol properties during a huge transatlantic smoke transport event using Aeolus observations in synergy with multi-platform data.	<i>Atmos Chem Phys</i>	Spaceborne HSRL case study with trajectory modelling	Aerosol extinction and backscatter from ALADIN; AOD	Global / multi-region
Embassy monitor shutdown, 46 cities	Air pollution after the shutdown of US embassy monitoring.	<i>Proc Natl Acad Sci U S A</i>	Matched difference-in-differences, satellite AOD	Real-time satellite aerosol optical depth, calibrated to pre-shutdown embassy PM _{2.5}	Global / multi-region
Feature engineering without co-location	Machine Learning-Based Calibration of Low-Cost PM _{2.5} Sensors Using Location-Specific Environmental Covariates and Feature Engineering Strategies.	<i>Chemosensors</i>	Ablation study, time-based validation	Low-cost PM _{2.5} sensor in an outdoor residential setting	Turkiye
PurpleAir Raleigh and Delhi	Evaluating and Calibrating Low-Cost Air Quality Sensors in Contrasting Aerosol Regimes.	<i>Aerosol Air Qual Res</i>	Co-location calibration campaign	PurpleAir PA-II PM _{2.5} against Federal Equivalent Method reference	United States; India
sMCPC supersaturation model	Performance modeling of a small mixing-type condensation particle counter.	<i>Atmos Meas Tech</i>	Numerical / CFD instrument model	Particle number, critical activation diameter Dp,50	Not applicable (laboratory / numerical)
Exposure assessment & modelling					
Monte Carlo wind erosion PM, Sardinia	Refinement and Quantitative Evaluation of a Monte Carlo Model for Wind-Driven PM Emissions from Industrial Granular Materials.	<i>Atmosphere</i>	Physical emission model, numerical sensitivity analysis	PM emission mass from saltation-driven sandblasting of granular surfaces	Europe
Portland segregation typologies, 342 tracts	Beyond redlining: Gentrification, displacement, disadvantages, and exclusivity predict urban environmental and health inequities.	<i>Proc Natl Acad Sci U S A</i>	Cross-sectional typology construction, PCA and rank correlation	Tract-level pollution exposure among 104 environmental, hazard and health variables	United States
Savannah and boreal fire vapour ageing	Organic vapors from Savannah and European Boreal fire emissions: insights from photochemical and dark aging.	<i>Atmos Chem Phys</i>	Controlled burn plus 29 m ³ chamber ageing, PTR-MS	Organic vapour emission factors, SOA precursors	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Southern Hemisphere fire trend 2003-2022	Characteristics and trend of the wild-fire activity in the Southern Hemisphere (2003-2022).	<i>Sci Total Environ</i>	Satellite time-series analysis	MODIS Active Fire detections as a proxy for biomass-burning PM source strength	Global / multi-region
Burden, policy & mitigation					
Central China BenMAP 2017-2024	Assessment of the spatiotemporal evolution of PM _{2.5} and its health and economic effects in Central China based on spatial statistics and BenMAP.	<i>Environ Monit Assess</i>	Spatial statistics plus BenMAP health impact assessment	Ground monitors and the Tracking Air Pollution in China (TAP) reanalysis, 2017-2024	China
PM stroke burden GBD 1990-2021	Ambient particulate matter pollution and stroke burden.	<i>Med Gas Res</i>	Descriptive burden analysis, GBD 2021	GBD ambient particulate matter pollution risk-factor exposure	Global / multi-region

Provenance and method

Entry window 2026-09-14 → 2026-09-14, contiguous with the 13 September issue; no gap. Built on 15 September under the standing rule that the current day is not opened before 22:00 local. **Sources.** PubMed [Date - Entry], run as two separate queries — health axis and instrumentation axis — through both the local harvester and the connector: 8 unique candidates, of which the instrumentation query's 2 were already in the health set. Registry-wide deposits for this entry date: 2,375 (13 September: 4,593). Europe PMC CREATION_DATE 3. Crossref windowed by ISSN across 25 tracked non-PubMed journals: 54 deposits, 14 in scope. arXiv relevance query returned 60 records, none inside the window. OpenAlex returned HTTP 429 and contributed nothing, as on every recent run. Consensus sweep on low-cost sensor calibration returned 10, of which 4 were net-new by title and 2 of those were then removed by the DOI gate as previously shipped under rewritten titles. **Screening.** 16 shipped, 11 rejected (9 out of scope, 2 duplicates), 6 held — DOIs verify but no abstract exists in Crossref, Europe PMC, Scholar Gateway or Consensus, and these are kept out of seen.json so a later issue can carry them. **Trial watch.** One registry record updated in window (NCT07817836, University of Pavia, urban air quality and work stress vs. inflammatory cytokines and DNA methylation, $n = 151$, observational, completed); accumulated to trials.json and surfaced in the weekly rollup, not here. **Labels.** Subtopic, design, particle-metric and geography are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Sixteen new design labels were added to the classifier in the same edit that wrote them onto records. Effect estimates are transcribed verbatim from abstracts. All DOI links resolve to the publisher of record except 10.4103/mgr.MEDGASRES-D-25-00188, which PubMed reports but which is not yet deposited with Crossref or Handle. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.